

When Birds Bow: An Interactive Audio-Visual Installation for Suona Culture via Collaborative Tangible User Interface

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Abstract

With the advancement of embodied interaction technologies, traditional musical instruments are increasingly viewed as interactive media for digital cultural expression. However, the symbolic dimensions of lesser-known traditional instruments such as the Suona, often remain difficult to externalize through conventional auditory-focused interfaces. This paper presents When Birds Bow, an interactive audio-visual installation that reimagines collaborative tangible interaction with the Suona into a narrative of birds offering feathers and the phoenix's rebirth. By mediating cultural translation through embodied, multisensory engagement, the system fosters collective participation and cultural resonance. Preliminary results suggest that this metaphor-driven interaction model enhances user engagement and deepens understanding of the Suona's symbolic significance, offering a new framework for reimagining traditional musical interfaces as vessels for cultural storytelling.

CCS Concepts

• Human-centered computing → interaction design.

Keywords

Suona Culture, Tangible User Interface, Collaborative Interaction, Cultural Preservation, Interactive Design

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1 INTRODUCTION

The Suona, a traditional Chinese wind instrument known for its ceremonial use, serves as a central role in folk rituals. Despite its cultural value, the Suona faces challenges in contemporary transmission, as younger generations become increasingly removed from the traditional ritual contexts. While existing digitization efforts

often focus on musical transcription, they rarely address the instrument's embodied ritual meanings. Moreover, interactive media seldom explore how traditional instruments like the Suona can be reinterpreted as expressive interfaces that communicate cultural metaphors beyond sound alone. Inspired by Hundred Birds Paying Homage to the Phoenix (Bai Niao Chao Feng), a canonical Suona piece known for imitating bird calls through pitch modulation, our installation reimagines the Suona's fingering interface as a tangible input system, where each gesture activates corresponding bird calls and animations via MIDI-translated notes, constructing a narrative ecosystem that culminates in the rebirth of the phoenix. To enable embodied interaction with the Suona installation, we combine light-responsive sensing and real-time audio-visual feedback, allowing individual and collaborative exploration of its communal symbolic meaning through tangible performance. In doing so, this work acts as a medium of cultural translation, transforming traditional instrumental interaction into a participatory ritual that evokes collective memory and cultural continuity.

2 RELATED WORK

The digital preservation of intangible cultural heritage (ICH) has become increasingly important, particularly as traditional Chinese ritual instruments fade from public awareness amid the decline of ritual practices. Prior work has explored the digitization of Chinese musical heritage. For example, Gong et al. introduced ChMusic, a dataset for Chinese instrument classification [Gong et al. 2021], and Chen et al. compiled Manchu music resources for AI-driven ethnomusicology [Chen et al. 2024]. While such efforts emphasize transcription and auditory fidelity, few explore how embodied, symbolic interaction, such as through tangible interfaces, can deepen cultural engagement with musical instruments. Recent studies have shown that tangible user interfaces (TUIs) can foster deeper cultural engagement through physical interaction and multisensory storytelling. For example, TUIs have been used to enrich educational heritage experiences [Hulusic et al. 2023] and to support ritual and spiritual expression through materially grounded design [Markum et al. 2024]. However, their integration into the context of traditional Chinese instruments like the Suona remains limited, particularly in forms that emphasize collaboration or ceremonial expression.

3 DESIGN AND IMPLEMENTATION

In the When Birds Bow project, participants engage in musical rituals using an interactive installation modeled after an oversized Suona, as illustrated in Figure 2. Guided by an on-site instruction panel, visitors collaboratively perform specific finger gestures on the Suona surface. As the ritual unfolds, these responses culminate

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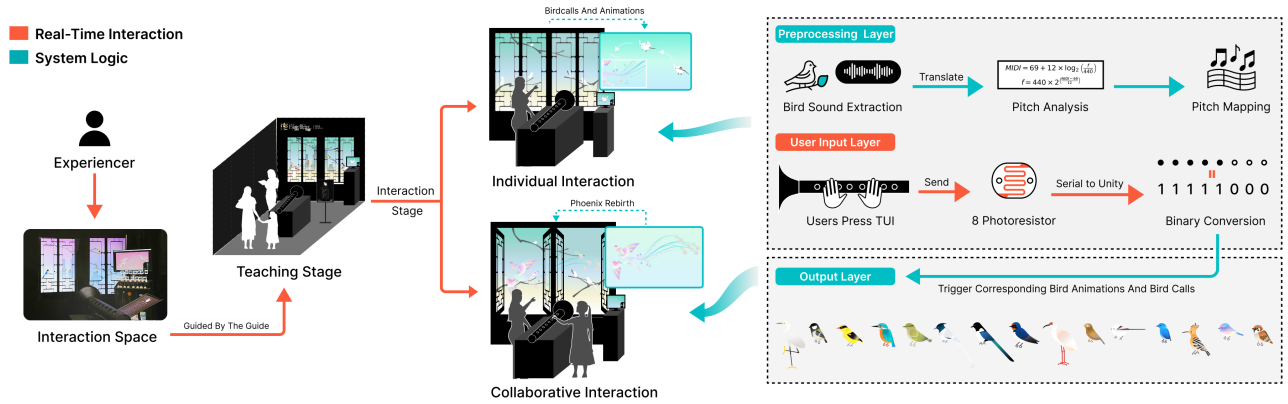


Figure 1: System Overview. The system integrates collaborative tangible interaction and multisensory feedback.

in the expressive rebirth of the phoenix, allowing participants to collectively revisit traditional myth and its philosophical meanings.

3.1 Design Concept

To evoke cultural resonance around the Suona, *When Birds Bow* reinterprets instrumental interaction as a symbolic, collaborative ritual. This multisensory transformation fosters embodied engagement and reframes the Suona as a dynamic cultural interface in contemporary contexts.

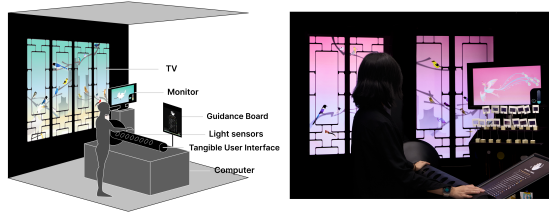


Figure 2: Installation layout and site situation.

3.2 Collaborative and Tangible Interaction

Utilizing the Suona Installation, our system turns musical performance into a collaborative ritual. Participants press specific tone holes on the Suona, each triggering a bird call and advancing a progress bar toward the samsara. Our design integrates light sensors beneath the Suona’s surface to detect these inputs and transmit them to Unity, enabling embodied interaction. The number of tone holes required to activate a bird call ranges from two to eight, which encourages participants to collaborate physically, fostering shared ritual engagement and cultural immersion.

3.3 Symbolic Audio-Visual Rituals

Notably, we harness a symbolic audio-visual workflow to translate Suona melodies into a culturally resonant experience, as shown in Figure 1. Our approach identifies 16 native bird species from Bai Niao Chao Feng, sourcing each call from archives and mapping it to a D-major Suona note via MIDI pitch conversion. Visually, we

emphasize the distinct features of each bird to evoke visual clarity and cultural familiarity.

4 EVALUATION

During on-site deployment, over 50 participants interacted with the installation. Informal feedback indicated that the multisensory interaction evoked cultural resonance and encouraged collaborative ritual participation, even among unfamiliar users—supporting the project’s cultural goals.

5 CONCLUSION

As we explore the ceremonial dimensions of Suona heritage, the integration of a tangible user interface in *When Birds Bow* represents a key innovation in cultural interaction. This installation invites the public to engage in collaborative rituals that translate traditional melodies into multisensory narratives. By lowering entry barriers and embedding cultural meaning in physical interaction, our system fosters renewed appreciation for the Suona. Future work will extend its educational reach in intangible heritage storytelling.

Acknowledgments

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